Jordan (Jay) Bell Compaan

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GitHub | Portfolio | Hugging Face

ABOUT ME

I am an AI researcher and full-stack developer with a First-Class Honours degree in Computer Science from Kingston University London, currently completing an MSc in Artificial Intelligence. My work is driven by turning questions into practical, well-evidenced answers, whether through research prototypes, software systems, or applied machine-learning pipelines.
I have led technical development for KURE-x-Tech's LunaNet project, a collaboration between the KURE and Tech Societies developed for NASA's LunaNet programme, and have over six years of freelance experience delivering software, 3D assets, and IT-support services. Across academic and independent projects, I combine strong software-engineering practice with rigorous evaluation, reproducibility, and responsible data governance. My recent work includes PrERT-CNM, a privacy-risk quantification framework; MetaboGuard, a longitudinal pan-cancer clinical-research model; BraCNIS, a brain-inspired continual-learning research project; and FakeNews, a supervised misinformation-detection research project. My work seeks to extend existing technical foundations to broaden the scope for innovation and development.

RESEARCH INTERESTS

Applied artificial intelligence and machine learning; privacy-enhancing technologies; trustworthy and interpretable AI; continual learning; clinical and biomedical prediction modelling; research software engineering.

EDUCATION

Kingston University London **2025–2026**
MSc Artificial Intelligence *London, UK*

- MSc research project: **PrERT-CNM**, an AI-driven privacy-risk quantification framework, supervised by Dr Razi Arshad.

  Developed a PrivBERT-based clause-classification pipeline with a Bayesian risk-scoring layer.

  Achieved test macro-F1 of 0.869, accuracy of 0.954, and expected calibration error of 0.0525.

Kingston University London **2021–2025**
BSc (Hons) Computer Science, First Class Honours *London, UK*

- Completed a foundation year in 2021 before progressing to the undergraduate degree.
- Developed strong foundations in software engineering, algorithms, databases, machine learning, and full-stack web development.

RESEARCH AND TECHNICAL PROJECTS

PrERT-CNM: AI Privacy-Risk Quantification Framework **2026**
MSc Research Project *Kingston University London*

- Designed and implemented an end-to-end framework for identifying and quantifying privacy risks in textual policy and regulatory content.
- Used PrivBERT for privacy-clause classification and integrated Bayesian modelling to produce interpretable, calibrated risk scores.
- Built evaluation and reporting workflows covering accuracy, macro-F1, calibration, and uncertainty-aware scoring.
- Deployed a public demonstration through Hugging Face Spaces and GitHub Pages.

- *Repository:* github.com/JayNightmare/PrERT-CNM-v4

KERI Research Project **2026**
Research Assistant *KERI Research*

- Developed a research-only platform for longitudinal clinical-data analysis, combining a Next.js/Vercel frontend with a Hugging Face-compatible Python modelling worker.
- Implemented modules for data ingestion, cohort definition, trajectory analysis, feature engineering, model training, calibration, evaluation, and reporting.
- Applied governance-focused development practices, including synthetic-data-only prototyping, provenance controls, model cards, and non-diagnostic disclaimers.

  Repository: github.com/KERI-Research
- Built a prototype for predicting possible cancer or diabetes development from longitudinal patient records using clinical and demographic features.
- Completed a 500-patient ordinary-incidence dataset and findings report.
- Implemented an auditable synthetic-prototype pathway with real-data authorisation gates and clear separation of simulation-only artefacts.

FakeNews: Misinformation Detection Research **2026**
Research Assistant *Kingston University*

- Developed a misinformation-detection research project under the supervision of Dr Peiling Yi.
- Created and released the **Misinformation Dataset**, a curated dataset for training and evaluating misinformation-classification models.
- Designed, trained, and evaluated the FakeNews classification model, releasing both the model and dataset through Hugging Face.
- *Repository:* github.com/JayNightmare/FakeNews
  Model: huggingface.co/JayNightmare/FakeNews
  Dataset: huggingface.co/datasets/JayNightmare/Misinformation-Dataset

BraCNIS: Brain-Inspired Continual Learning **2026**
Independent Research Project *Neuromodulatory Continual Learning*

- Developed BraCNIS, a research project exploring continual learning using neuromodulatory internal-state variables.
- Investigated approaches to reducing catastrophic forgetting and dependence on repeated retraining.
- Produced experimental results and supporting material for a work-in-progress research preprint.
- *Repository:* github.com/JayNightmare/BraCNIS

PROFESSIONAL EXPERIENCE

KURE-x-Tech – LunaNet **April–September 2026**
Lead Developer *Remote / London, UK*

- Led technical development for LunaNet, a KURE and Tech Societies collaboration developing a software reference implementation for NASA's LunaNet programme.
- Contributed to system architecture, implementation, documentation, testing, code quality, and development workflows.

Fiverr – Freelance Developer **2019–2025**
Self-Employed *Remote*

- Delivered software development, 3D game assets, and IT-support services to clients for over six years.
- Developed experience in client communication, requirements gathering, iterative delivery, and practical full-stack development.

Youngs Pub – The Victora, Surbiton **2024–Present**
Bartender *London, UK*

- Provided customer service in a fast-paced hospitality environment.
- Developed communication, teamwork, reliability, and customer-facing problem-solving skills.
- Trained in Service Master, Cask Ale and Beer, and Leadership Qualifications.

SELECTED SOFTWARE PROJECTS

- **PhunParty** — Collaboration with a friend to build a party game. Repository
- **FilmReel** — TypeScript web application for film discovery. Repository
- **Monolith** — Marketing-analytics application using Gemini 2.5 Flash to generate audience personas and insights from video content. Repository
- **WeSDK** — Revamped Luffa JavaScript SDK with Promise-first APIs, TypeScript support, runtime capability checks, and legacy compatibility. Repository
- **Mars Client** — Custom-made Minecraft Client which hooks into several endpoints for an automated setup. Repository

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript, SQL, Rust (working knowledge)
AI/ML: Transformers, BERT/PrivBERT, Bayesian modelling, model evaluation, calibration, continual learning, NLP, computer vision
Web Development: React, Next.js, Vite, Tailwind CSS, REST APIs, Node.js
Research Engineering: Jupyter, experiment pipelines, ablation studies, model cards, reproducibility, data governance, research reporting
Tools and Deployment: Git, GitHub, GitHub Actions, Hugging Face Spaces, Vercel, Cloudflare, Neon, Docker

PUBLIC PROFILES

- **GitHub:** github.com/JayNightmare — 150+ repositories, including private projects; maintains seven public-facing repositories
- **Portfolio:** portfolio.nexusgit.info
- **Hugging Face:** huggingface.co/JayNightmare